

CyberTrust KSA

AI-Driven NCA Compliance Platform Developer Prototype & Technical Brief (Version 3.1)

1. Platform Brief for Technical Team

CyberTrust KSA is an AI-powered SaaS platform designed to automate and streamline National Cybersecurity Authority (NCA) compliance for organizations in Saudi Arabia. The v3.1 architecture introduces a multi-framework capability, allowing organizations to assess against NCA standards (NCNICC-1:2025 or ECC-2:2024) alongside vendor-specific requirements such as SABIC CyberTrust and Saudi Aramco CCC simultaneously.

2. Developer FAQ

Q: How does the new Smart Classification work? A: The AI Intake Questionnaire uses smart branching. Based on Sector (Section A), Size (Section B), and Vendor targets (Sections D & E), the AI determines the exact control sets required.

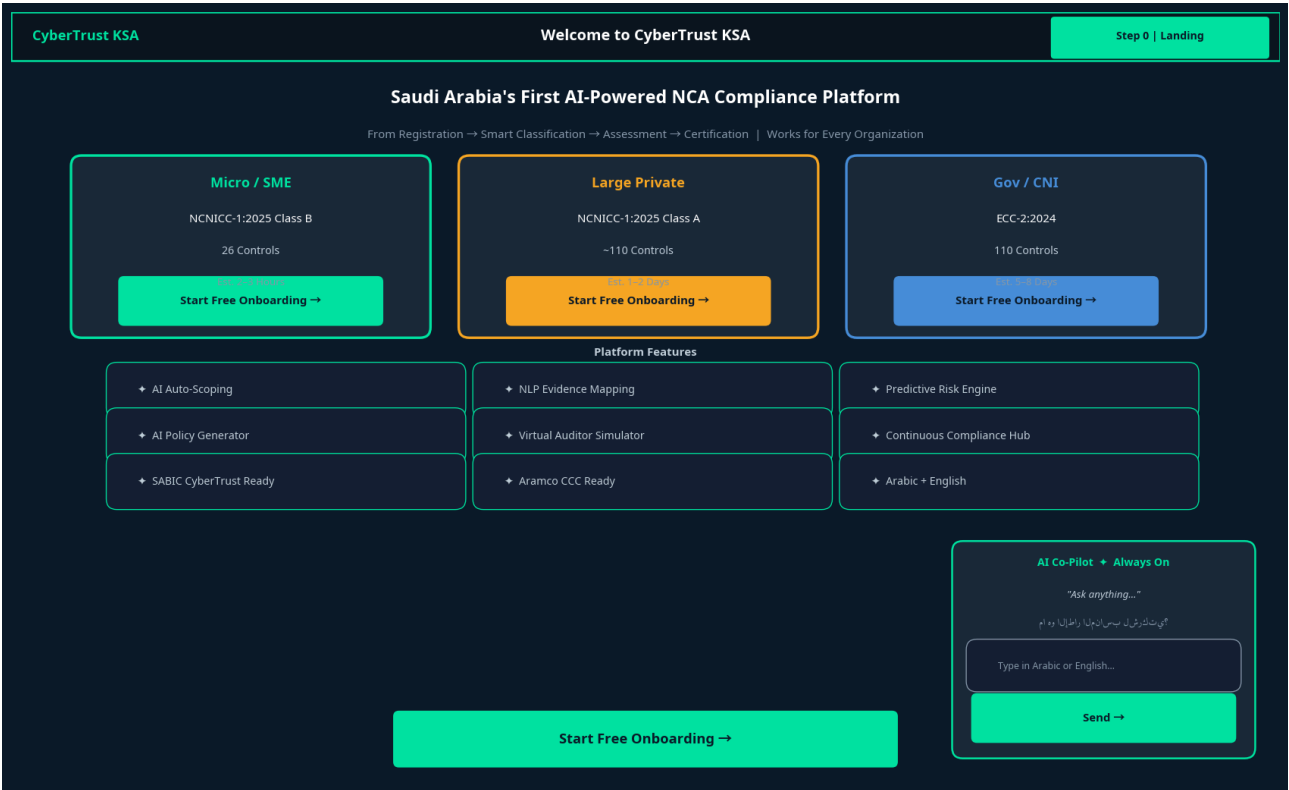
Q: How are multiple frameworks handled? A: The system merges control sets from all active frameworks into a unified `assessment_scope`. Shared controls (e.g., Access Control) are deduplicated and tagged with multiple framework badges.

Q: What happens when a user selects SABIC or Aramco? A: Selecting SABIC in Q A3 renders Section D (SABIC CyberTrust Classification). Selecting Aramco renders Section E (Aramco CCC Classification). The results dictate which specific vendor controls are activated.

3. Prototype Walkthrough & Scenarios (Multi-Framework Edition)

Phase 0: Landing Page

Scenario: User lands on the platform. AI features and 3-tier framework pricing are displayed.



Phase 0: Registration

Scenario: User inputs organization details. Data is encrypted and stored in KSA for data residency compliance.

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Organization Registration

Step 1 | Registration

Create Your Organization Profile

This information is used by the AI to pre-fill your classification and identify your applicable NCA framework

Organization Details

Organization Name *

Al-Rashidi Technology Solutions LLC

Commercial Registration No. *

1010XXXXXX

Primary Industry / Sector *

Information Technology & Software

Organization Type *

Private Limited Company (LLC)

Contact & Location

City / Region *

Riyadh, Saudi Arabia

Website (Optional)

www.alrashidi-tech.com

Primary Contact Name *

Ahmed Al-Rashidi

Contact Email *

ahmed@alrashidi-tech.com

Your data is encrypted and stored in Saudi Arabia (KSA data residency). Used only for NCA compliance assessment.

Next: Start Classification →

Phase 1: Smart Classification — Section A (Sector & Vendor Path)

Scenario: User selects “Private Sector” and indicates they are targeting the SABIC CyberTrust program. The AI Co-Pilot notes it will collect SABIC-specific data.

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Smart Classification — Section A: Sector & Vendor Path

Step 2A | Classification

A: Sector

B: Size

C: Tech Scope

D: SABIC

E: Aramco

Summary

Section A: Organizational Identity & Sector

The AI uses your answers to determine which NCA framework applies and which vendor programs to activate.

Q A1 — What is your organization's primary sector?

This is the primary routing question. Government/CNI organizations are immediately routed to ECC-2:2024 (110 controls).

1. Government / Ministry
→ Route to ECC-2:2024 (110 controls)

2. Private Sector

→ Proceed to A2

3. Semi-Government / State-Owned
→ Flag for AI review

Q A2 — Does your organization own, operate, or manage Critical National Infrastructure (CNI)?

CNI sectors: Energy, Water, Telecom, Finance, Healthcare, Transport, Government IT. If Yes → ECC-2:2024 mandatory.

1. Yes
→ Route to ECC-2:2024

2. No

→ Proceed to A3

3. Not Sure
→ AI asks follow-up industry questions

Q A3 — NEW in v2.1 Vendor/supplier programs are you targeting? (Multi-select allowed)

This activates parallel framework assessment alongside NCA. Select all that apply.

Saudi Aramco
(Vendor/Supplier)

→ Activates Section E
(CCC Classification)

SABIC
(Vendor/Supplier)

→ Activates Section D
(CyberTrust Classification)

NCA Compliance Only
(No vendor program)

→ Skip to Section B
(Size & Revenue only)

AI Co-Pilot:

"You selected SABIC as a vendor target. I will now collect SABIC-specific classification data (service type, data access, cloud/OT usage) alongside your NCA classification."

Developer Note:

Store vendor selections in session state. Conditionally render Section D (SABIC) and/or Section E (Aramco) based on A3 selections.

Next: Section B →

Phase 1: Smart Classification — Section B (Size & Revenue)

Scenario: User inputs 6-249 employees and 3M-200M SAR revenue. The AI classifies them as NCNICC-1:2025 Class B.

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A: Sector ✓

B: Size

C: Tech Scope

D: SABIC

E: Aramco

Summary

Step 2B | Classification

Section B: Size & Revenue Classification

Determines whether you fall under NCNICC-1:2025 Class A (Large) or Class B (SME/Micro). Only applies to Private Sector organizations.

Q B1 — How many full-time employees currently work at your organization?

Threshold: 250+ employees = NCNICC-1:2025 Class A | 6-249 = Class B | 1-5 = Class B (micro, recommended compliance)

1-5 Employees (Micro)
→ NCNICC-1:2025 Class B
26 baseline controls

6-249 Employees (SME)

→ NCNICC-1:2025 Class B
26 baseline controls

250+ Employees (Large)
→ NCNICC-1:2025 Class A
~110 controls (ECC depth)

Q B2 — What is your organization's estimated annual revenue?

Threshold: >200M SAR = NCNICC-1:2025 Class A | 3M-200M SAR = Class B | <3M SAR = Class B (recommended)

Less than 3M SAR (Micro)
→ NCNICC-1:2025 Class B
(Recommended compliance)

3M - 200M SAR (SME)

→ NCNICC-1:2025 Class B
(Mandatory compliance)

More than 200M SAR (Large)
→ NCNICC-1:2025 Class A
(Mandatory, full scope)

AI Co-Pilot:

Developer Note:

Override Rule:

"Based on your answers (6-249 employees, 3M-200M SAR revenue), you are classified as NCNICC-1:2025 Class B. You will be assessed against 26 baseline controls."

If B1=3 OR B2=3 → assign Class A. Else assign Class B. Store result in org_profile.classification. This drives the control set loaded in Phase 3.

If EITHER employees OR revenue hits the Class A threshold, Class A applies — even if the other metric is below threshold.

Next: Section C →

Phase 1: Smart Classification — Section C (Technical Scope)

Scenario: User indicates they do not host custom web applications. The AI auto-marks Subdomain 2-15 as N/A and generates a justification.

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Smart Classification — Section C: Technical Scope Refinement

Step 2C | Classification

A: Sector ✓B: Size ✓C: Tech ScopeD: SABICE: AramcoSummary

Section C: Technical Scope Refinement — N/A Auto-Marker

These 4 questions auto-remove non-applicable subdomains from your assessment. Answers apply to ALL active frameworks (NCA + SABIC + Aramco).

Q C1 — Do you use Cloud Computing services?

(e.g., AWS, Azure, STC Cloud, SaaS applications)

✓ Yes
Cloud subdomain stays ACTIVE for NCA + SABIC

✗ No
Domain 4 (Cloud) → Auto-marked N/A
AI generates justification: 'No cloud services in scope'

Q C2 — Do you outsource IT or cybersecurity to third-party vendors?

(e.g., managed services, external IT support, security operations)

✓ Yes
Third-Party subdomain stays ACTIVE

✗ No
Domain 4 (Third-Party) → Auto-marked N/A
AI generates justification: 'No third-party IT services'

Q C3 — Do employees access company data using mobile devices?

(MDM/BYOD — company or personal phones/tablets)

✓ Yes
Subdomain 2-6 (Mobile/MDM) stays ACTIVE

✗ No
Subdomain 2-6 (Mobile) → Auto-marked N/A
AI generates justification: 'No mobile device access'

Q C4 — Do you develop or host custom web applications?

(Internal portals, customer-facing apps, APIs)

AI Co-Pilot:
✓ Yes
Developer Note: Web Apps stays #6TW/ach 'No' answer, call: markControlNA(subdomain_id, auto_justification_text). Store in assessment table for Phase 5. Subdomain 2-15 (Web Apps) → Auto-marked N/A. User selected 'No' → marked N/A.

✗ No
Subdomain 2-15 (Web Apps) → Auto-marked N/A
User selected 'No' → marked N/A.

Next: Section D (SABIC) →

Phase 1: Smart Classification — Section D (SABIC CyberTrust)

Scenario: (Conditional) User selects GR and SM service types and confirms data access. The AI activates the corresponding SABIC controls.

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Smart Classification — Section D: SABIC CyberTrust (Conditional)

Step 2D | SABIC Classification

A: Sector ✓B: Size ✓C: Tech Scope ✓D: SABICE: AramcoSummary

This section appears ONLY because you selected 'SABIC' in Question A3. It runs in parallel with your NCA assessment.

Section D: SABIC CyberTrust Classification

Determines which SABIC CyberTrust control categories apply based on your service/product type and data access level.

Q D1 — What type of services/products will you provide to SABIC? (Multi-select)

Select all categories that describe your engagement. Each selection activates the corresponding CyberTrust control set.

GR General Requirement
Access to SABIC data

NC Network Connectivity
Direct SABIC network access

CCS Cloud Computing Services
Cloud solutions for SABIC

OMS Outsourcing & Managed Services
Managed IT/security services

CS Consultancy Services
Handling classified data

SM Software Management/Dev
Software solutions/development

OT OT/ICS Products & Services
Operational technology

Q D2 — Direct access to SABIC network/systems?

Triggers higher Network Connectivity (NC) controls

✓ Yes
→ NC controls activated at higher level

No

Q D3 — Handle/process SABIC classified or sensitive data?

Triggers stricter data protection and access control requirements

✓ Yes
→ GR + SM data protection controls activated

No

AI Co-Pilot:

Great. I have classified your SABIC engagement as GR + SM (General Requirement + Software Management). I will now collect SABIC-specific evidence alongside your NCA controls.

Developer Note:

Store SABIC classification in vendor_profiles table: (vendor: 'SABIC', categories: ['GR','SM'], network_access: true, data_access: true).

Control Mapping:

Load SABIC CyberTrust control set filtered by selected categories. Merge with NCA controls in unified assessment_scope. Shared controls (e.g., access control, encryption) shown once with dual-framework tags.

Conditional Logic:

Section D only renders if A3 includes 'SABIC'. If user skips or deselects SABIC, all Section D data is cleared from session.

Next: Section E (Aramco) →

Phase 1: Smart Classification — Section E (Aramco CCC)

Scenario: (Conditional) User selects IT Services and standard CCC path. The AI adds Aramco controls to the scope.

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Smart Classification — Section E: Aramco CCC (Optional)

Step 2E | Aramco Classification

A: Sector ✓

B: Size ✓

C: Tech Scope ✓

D: SABIC ✓

E: Aramco

Summary

✖ This section appears ONLY if 'Saudi Aramco' was selected in Question A3. It is optional and can be added later.

Section E: Aramco CCC Classification (Contractor Cybersecurity Compliance)

Determines your Aramco CCC path (standard vs. CCC+) based on commodity category, facility type, and IKTVA status.

Q E1 — Which Aramco commodity/category are you targeting? (Select main category)

Helps determine CCC vs. CCC+ path. CCC+ is required for higher-risk categories (OT, critical infrastructure supply).

✓ IT Services & Software

○ Engineering & Consulting

○ OT/ICS & Industrial

○ Construction & Maintenance

○ Manufacturing & Supply

○ Logistics & Transport

○ Healthcare & Safety

○ Financial & Legal Services

○ Other / General

Q E2 — Do you have manufacturing facilities or plants that will supply Aramco?

Manufacturing/plant facilities typically trigger CCC+ (on-site audit requirement).

Yes → CCC+ Path

✓ No → CCC Path

→ Standard CCC path applies (no on-site audit)

Q E3 — What is your current IKTVA / local content status?

IKTVA (In-Kingdom Total Value Add) status affects Aramco readiness scoring.

Not Started

In Progress

Certified

→ Used for Aramco readiness scoring in Pre-Audit phase

AI Co-Pilot:

"You are targeting Aramco in the IT Services & Software category. Based on your answers, you are on the standard CCC path (no on-site audit required). I have added Aramco CCC controls to your multi-framework dashboard."

Developer Note:

Store Aramco profile: (vendor: 'Aramco', commodity: 'IT Services', ccc_path: 'standard', iktva: 'in_progress'). Load Aramco CCC control set and merge into unified assessment_scope.

CCC vs CCC+ Logic:

If E2 = Yes (manufacturing) OR commodity in [OT/ICS; Construction] → assign CCC+ path. CCC+ requires on-site audit flag in audit_requirements table.

Skip Option:

If user did not select Aramco in A3, this section is hidden entirely. User can 'Add Aramco Path Later' from the dashboard.

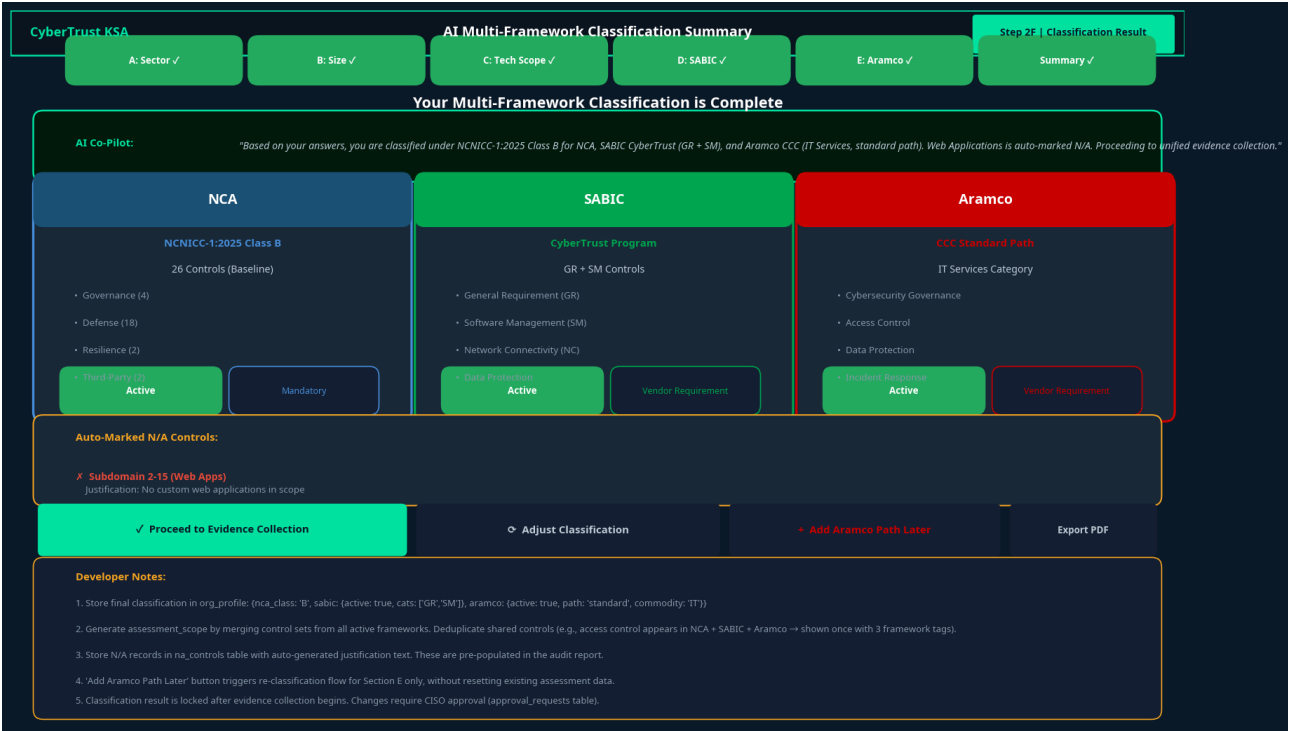
Parallel Frameworks:

The platform now maintains 3 parallel frameworks: NCA (NCNICC-1:2025 Class B) + SABIC CyberTrust (GR+SM) + Aramco CCC (IT Services, standard path).

Next: Summary →

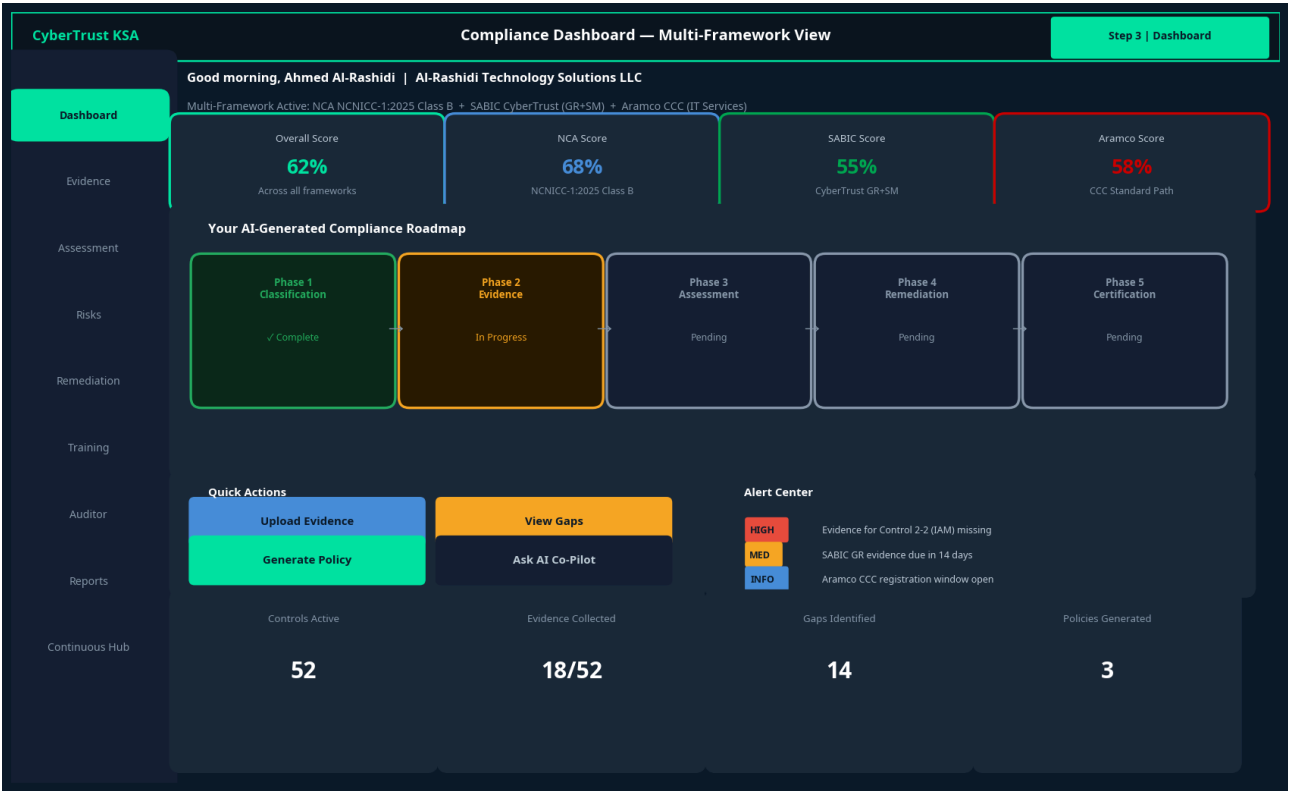
Phase 1: AI Classification Summary

Scenario: The AI presents the final multi-framework classification (NCA + SABIC + Aramco) and the auto-marked N/A controls before proceeding to evidence collection.



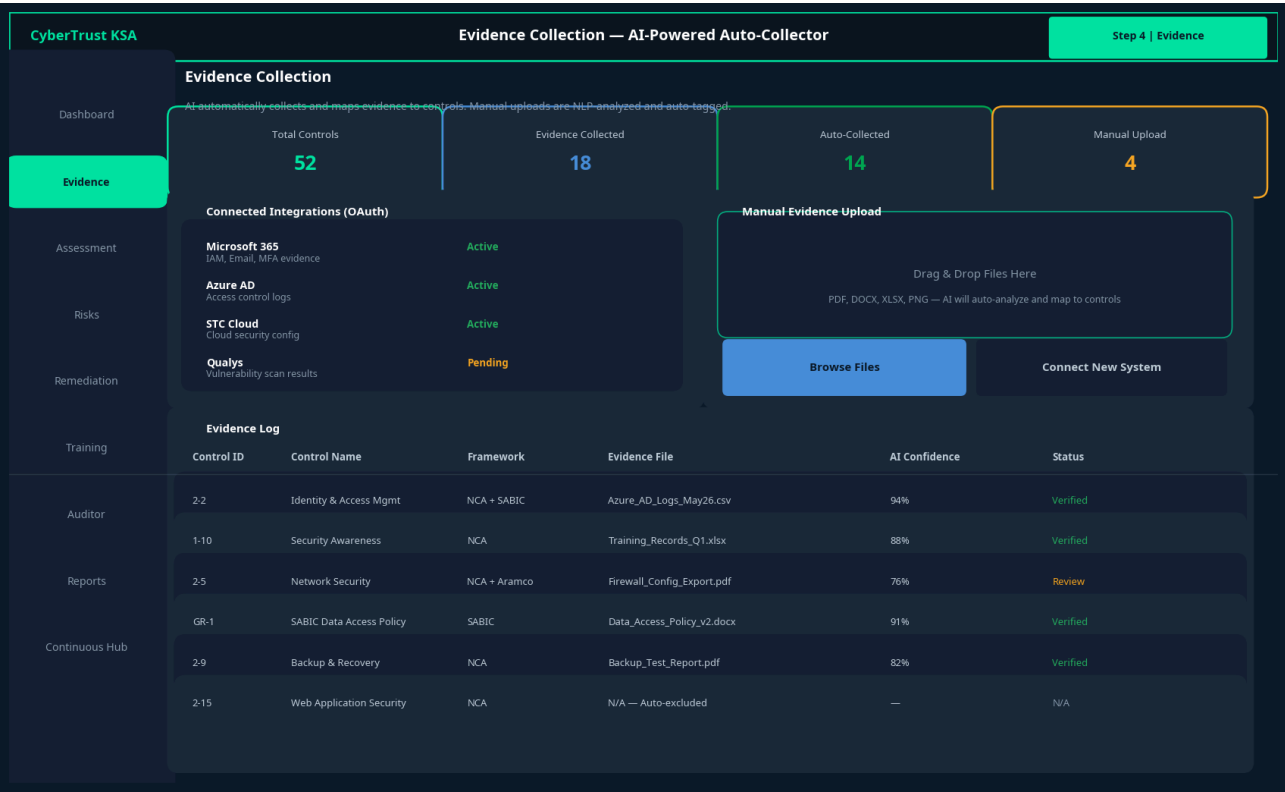
Phase 2: Compliance Dashboard

Scenario: Personalized dashboard showing overall score and individual scores for NCA, SABIC, and Aramco.



Phase 3: Evidence Collection

Scenario: Auto-collector via OAuth integrations. AI NLP parses manual uploads and maps them to controls across all active frameworks.



Phase 4: Control Assessment

Scenario: Unified control view. Shared controls show multiple framework tags. AI pre-fills status based on evidence.

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Control Assessment — AI Pre-Filled, Multi-Framework

Step 5 | Assessment

Dashboard

Evidence

Assessment

Risks

Remediation

Training

Auditor

Reports

Continuous Hub

Control Assessment

AI pre-fills compliance status based on collected evidence. Review and confirm each control. Multi-framework tags show which frameworks each control satisfies.

All Frameworks (52)

NCA Only (26)

SABIC Only (12)

Aramco Only (14)

Shared Controls (18)

Control ID	Control Name	Frameworks	AI Status	Evidence	AI Confidence	Action
1-4	Cybersecurity Roles & Responsibilities	NCA	Compliant	Job Descriptions v2.do...	89%	✓ Confirmed
2-2	Identity & Access Management	NCA SABIC	Aramco Compliant	Azure_AD_Logs.csv	94%	✓ Confirmed
2-5	Network Security	NCA Aramco	Partial	Firewall_Config.pdf	76%	Upload Evidence
GR-1	SABIC Data Access Policy	SABIC	Compliant	Data_Policy_v2.docx	91%	✓ Confirmed
2-9	Backup & Recovery	NCA	Compliant	Backup_Report.pdf	82%	✓ Confirmed
2-10	Vulnerability Management	NCA Aramco	Non-Compliant	No evidence uploaded	—	Generate Policy
SM-3	Secure Software Development	SABIC	Partial	SDLC_Policy.docx	68%	Upload Evidence
2-13	Incident Response	NCA SABIC	Aramco Non-Compliant	IRP not documented	—	Generate Policy
2-15	Web Application Security	NCA	N/A	Auto-excluded (no web ...	—	View Justification

Phase 5: Gap Analysis & Predictive Risk

Scenario: Risk Heatmap and Predictive Risk Engine showing probability of major findings for open gaps.

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Gap Analysis & Predictive Risk Engine

Step 6 | Gap Analysis

Dashboard

Evidence

Assessment

Risks

Remediation

Training

Auditor

Reports

Continuous Hub

Gap Analysis & Predictive Risk Engine

AI identifies compliance gaps and predicts the probability of a major auditor finding if gaps remain unresolved.

Critical Gaps3

High Gaps5

Medium Gaps6

Resolved8

Predictive Risk Engine

Probability of major finding if gap left open:

Incident Response (2-13)91%

Vulnerability Mgmt (2-10)78%

Secure SDLC (SM-3)62%

Risk Heatmap

Low

Med

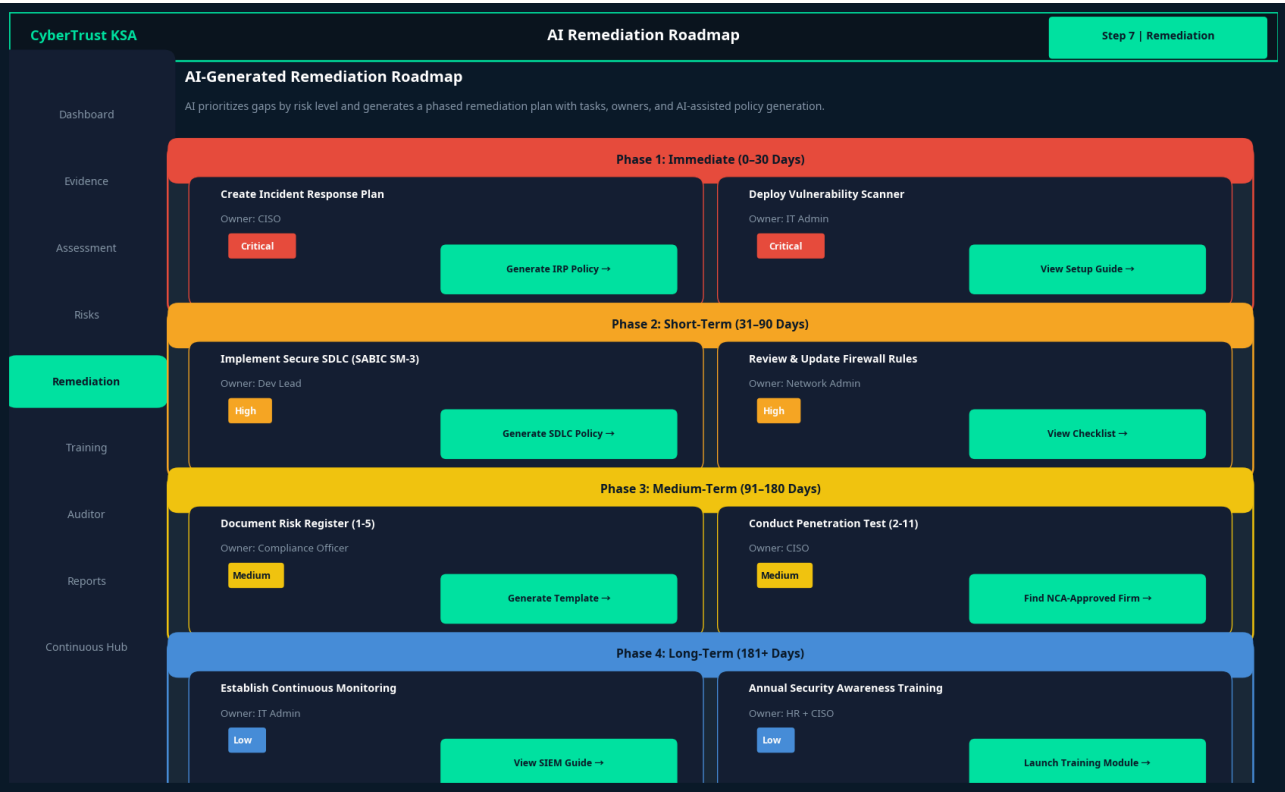
High

Prioritized Gap List

Gap ID	Control	Framework	Risk Level	Finding Probability	Recommended Action
G-001	Incident Response (2-13)	NCA+SABIC+Aramco	Critical	91%	Create IRP immediately
G-002	Vulnerability Mgmt (2-10)	NCA+Aramco	Critical	78%	Deploy Qualys scanner
G-003	Secure SDLC (SM-3)	SABIC	High	62%	Adopt OWASP SDLC framework
G-004	Network Segmentation (2.5)	NCA+Aramco	High	55%	Review firewall rules
G-005	Risk Register (1-5)	NCA	Medium	38%	Document risk register

Phase 6: Remediation Roadmap

Scenario: AI-generated 4-phase remediation plan with tasks, owners, and AI policy generation buttons.



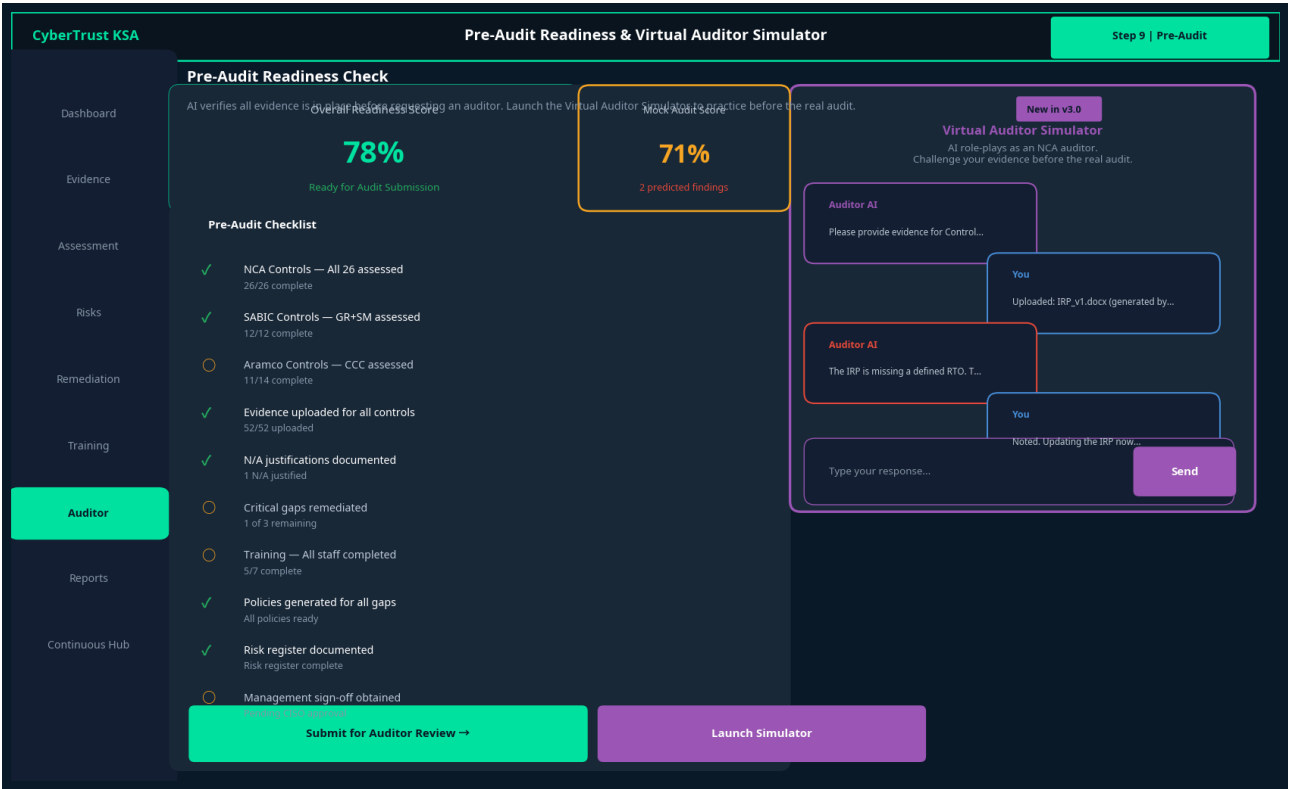
Phase 7: Training Module

Scenario: Personalized phishing simulation campaigns. Results feed directly into Domain 1-6 compliance score.



Phase 8: Pre-Audit & Simulator

Scenario: Readiness checklist across all frameworks. User can launch Virtual Auditor Simulator to role-play the audit.



Phase 9: Auditor Portal

Scenario: Auditor reviews evidence. AI highlights NLP findings. Auditor sets final verdict and submits report.

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Auditor Portal — Control Review & Findings

Step 10 | Auditor Review

My Cases

Control Review

Auditor Portal — NCA Approved Auditor

Case: Al-Rashidi Technology Solutions LLC | Framework: NCNICC-1:2025 Class B + SABIC CyberTrust | Readiness Score: 78%

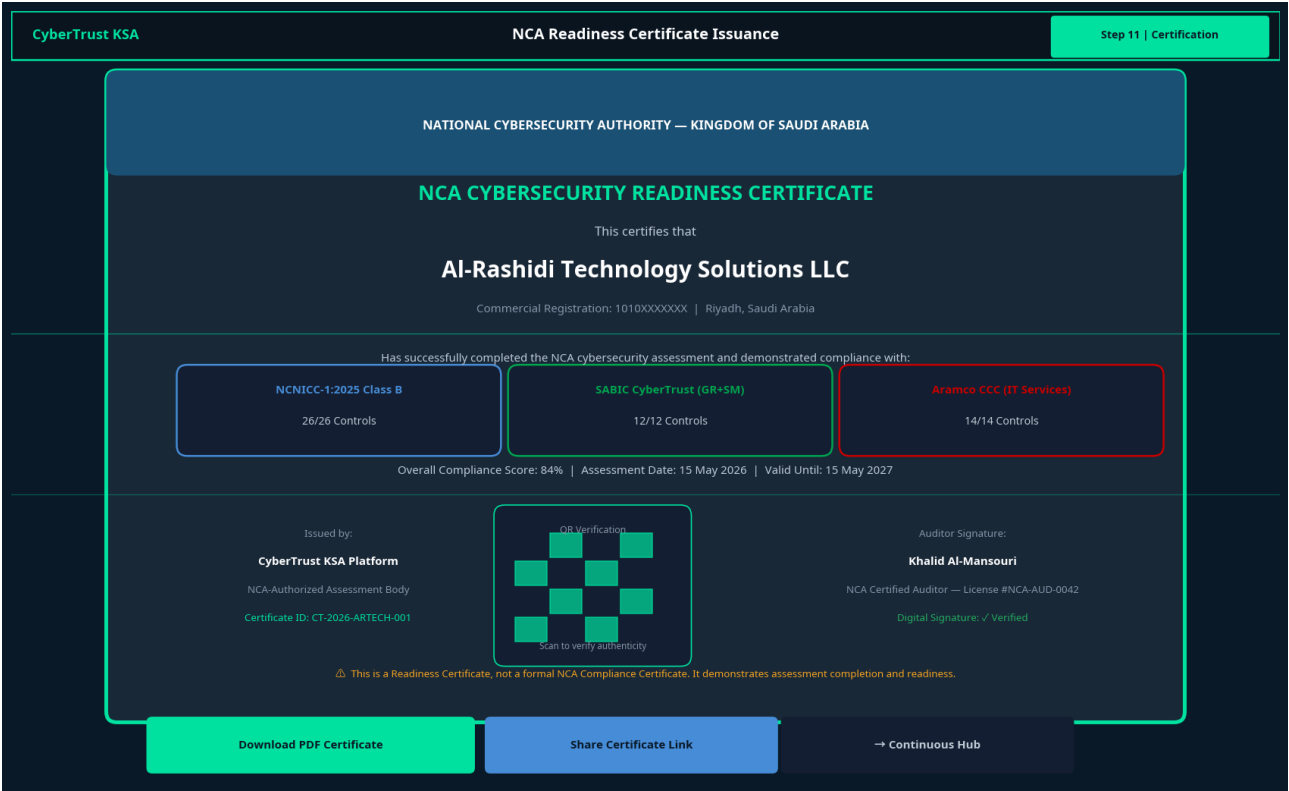
Control-by-Control Review

AI has pre-analyzed all evidence and provided a suggested verdict. Auditor can override with formal comments.

Control	Name	Framework	AI Verdict	Evidence	Auditor Verdict	Comments	
Documents	1-4	Roles & Responsibilities	NCA	Verified	Job Description...	Verified	—
	2-2	Identity & Access Mgmt	NCA+SABIC	Verified	Azure AD Logs	Verified	—
Findings	2-5	Network Security	NCA+Aramco	Conditional	Firewall Config	Conditional	Missing VLAN docs
	2-10	Vulnerability Mgmt	NCA+Aramco	Finding	No evidence	Finding	Critical gap — no ...
Submit Report	2-13	Incident Response	NCA+SABIC+Aramco	Finding	IRP v1.docx	Conditional	IRP missing RTO de...
	GR-1	SABIC Data Access Policy	SABIC	Verified	Policy v2.docx	Verified	—
	SM-3	Secure SDLC	SABIC	Conditional	SDLC Policy.doc...	Conditional	Needs code review ...
	2-15	Web App Security	NCA	N/A	Auto-excluded	N/A	Accepted justifica...
	Submit Findings & Report →						

Phase 10: Readiness Certificate

Scenario: Verifiable digital certificate showing completion of NCA, SABIC, and Aramco assessments with QR code.



Phase 10: Continuous Compliance Hub

Scenario: Live score updates daily. Evidence auto-refreshes. Alert center for drift and expiring evidence.

